

Opponent-Aware Q-Learning for Leduc Hold'em Poker

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Abstract—We investigate opponent-aware Q-learning for Leduc Hold'em poker — a tractable benchmark with a computable Nash equilibrium. Standard RL agents learn a fixed policy and do not adapt to opponent tendencies. We propose augmenting a Q-learning agent with a lightweight opponent model that tracks action-frequency statistics (bet, raise, fold rates) over recent hands and classifies the opponent into one of three behavioral styles: random, tight, or aggressive. The Q-learning agent's state representation is extended with this style indicator, enabling it to adapt its strategy accordingly. Baselines include a random agent, a rule-based agent, and a standard Q-learning agent. We evaluate using win rate, average reward, and learning curves.

Index Terms—reinforcement learning, poker, Leduc Hold'em, opponent modeling, Q-learning, imperfect information games

I. INTRODUCTION

Poker is a canonical benchmark for decision-making under uncertainty. Unlike perfect-information games such as Chess or Go, it requires reasoning over hidden state, managing multi-round strategy, and adapting to opponent behavior — all core challenges in reinforcement learning (RL).

Recent breakthroughs such as Libratus [1] and Pluribus [2] achieved superhuman performance in No-Limit Texas Hold'em (NLHE), but these systems primarily target Nash equilibrium strategies that are robust against any opponent rather than adaptive strategies that exploit specific tendencies. Leduc Hold'em [8], with only 6 cards and 2 betting rounds, preserves the core challenges of poker while remaining computationally tractable.

Most RL agents for poker assume a stationary opponent. In practice, opponents exhibit consistent behavioral patterns: some fold frequently (tight), others bet and raise aggressively, and others play randomly. An agent that detects and exploits these patterns should outperform a fixed-policy baseline. Our contributions are:

- A simple, frequency-based opponent style classifier requiring no recurrent networks or Bayesian inference.
- An opponent-aware Q-learning agent that conditions its policy on the inferred opponent style.
- A comparison against random, rule-based, and standard Q-learning baselines using win rate and average reward.

II. LITERATURE SURVEY

A. Counterfactual Regret Minimization (CFR)

CFR [5] is the foundational algorithm for solving extensive-form imperfect-information games. By iteratively minimizing counterfactual regret at each information set, it converges to a Nash equilibrium. Variants including CFR+ [6] and Monte Carlo CFR [7] improve practical convergence speed. RLCard provides a built-in CFR implementation for Leduc Hold'em, which we use directly as an approximate Nash reference agent.

B. Neural Fictitious Self-Play (NFSP)

NFSP [3] combines DQN-based RL with a supervised average-strategy network, provably converging to Nash equilibrium in two-player zero-sum games. While powerful, NFSP requires maintaining two separate networks and a reservoir buffer, making it complex to implement from scratch. We use RLCard's NFSP implementation as a reference point in our literature survey but simplify our own baseline to Q-learning.

C. Superhuman Agents in NLHE

DeepStack [4] introduced depth-limited solving with a learned value network, achieving human-expert level in heads-up NLHE. Libratus [1] combined abstraction, real-time solving, and opponent exploitation to defeat professional players. Pluribus [2] extended this to six-player NLHE. These systems target Nash equilibrium rather than explicit opponent adaptation, and their scale makes them unsuitable as direct baselines for our work.

D. Opponent Modeling in Poker

Early opponent modeling used Bayesian methods to maintain distributions over opponent hand ranges [8]. More recent work encodes opponent observations into neural networks using Mixture-of-Experts architectures to discover opponent strategy clusters [10]. Rahman et al. [11] proposed AMP3 for multi-player Texas Hold'em, feeding predicted opponent style features into an Actor-Critic agent. To our knowledge, no prior work has studied frequency-based opponent modeling specifically on Leduc Hold'em with Q-learning as the base agent.

III. BACKGROUND

A. Leduc Hold'em

Leduc Hold'em is a two-player poker game with a 6-card deck (Jack, Queen, King in two suits). Each player receives one private card and posts a fixed ante of 1 chip. Play proceeds over two betting rounds: pre-flop and post-flop (after one community card is revealed). Each round permits a maximum of two raises. The winner is determined by showdown rank (pairs beat high cards) or by the opponent folding.

B. Problem Formulation

We model Leduc Hold'em as a Partially Observable Markov Decision Process (POMDP), formally defined by the tuple (S, A, T, R, O) . The agent operates on its private observation vector o rather than the full game state s . RLCard encodes o as a fixed-length 36-dimensional binary vector, and the state space is small enough to support tabular Q-learning without function approximation.

IV. BASELINE IMPLEMENTATION

We use the RLCard toolkit [12] for all experiments, which provides the Leduc Hold'em environment as an approximate Nash utilities.

Random Agent. Selects uniformly at random. Serves as the lower-bound baseline.

Rule-Based Agent. A hand-coded strategy that plays a simple but non-trivial fixed strategy.

Q-Learning Agent. A tabular Q-learning agent that plays against a random opponent using ϵ -greedy exploration as the primary RL baseline and the proposed method.

V. PROPOSED METHOD: OPPONENT-AWARE Q-LEARNING

A. Motivation

The standard Q-learning agent treats the opponent as a fixed unknown part of the environment. Real opponents exhibit systematic behavioral patterns. An agent that detects these patterns and adapts its strategy should achieve higher win rates. We propose a minimal, practical mechanism to achieve this without recurrent networks or Bayesian inference.

B. Opponent Style Classification

We track three action frequency statistics over a sliding window of the last $W=20$ observed opponent actions: f_{bet} (call+check rate), f_{raise} (raise rate), f_{fold} (fold rate). The opponent is classified into one of three styles using simple thresholds:

- **Aggressive:** $f_{\text{raise}} > 0.45$
- **Tight:** $f_{\text{fold}} > 0.40$ and $f_{\text{raise}} < 0.05$
- **Random:** otherwise (default before W observations)

This requires no learned parameters and updates in $O(1)$ per step.

C. Augmented State Representation

The opponent-aware agent's state is augmented with the inferred style indicator $c \in \{\text{random, tight, aggressive}\}$: $s' = (s, c)$, where s is the standard RLCard observation vector. The Q-table is indexed by (s', a) , allowing the agent to learn distinct policies for each opponent style. No neural network or sequence model is required.

D. Training Against Fixed-Style Opponents

We train the opponent-aware agent in rotating blocks of 30 episodes against three scripted opponents: Random, Tight, and Aggressive. The classifier detects each switch, enabling the agent to populate the correct Q-table partition per style.

E. Evaluation

We evaluate on win rate and average reward per episode across 10,000 evaluation games, comparing all agents vs. all opponent types.

VI. RESULTS

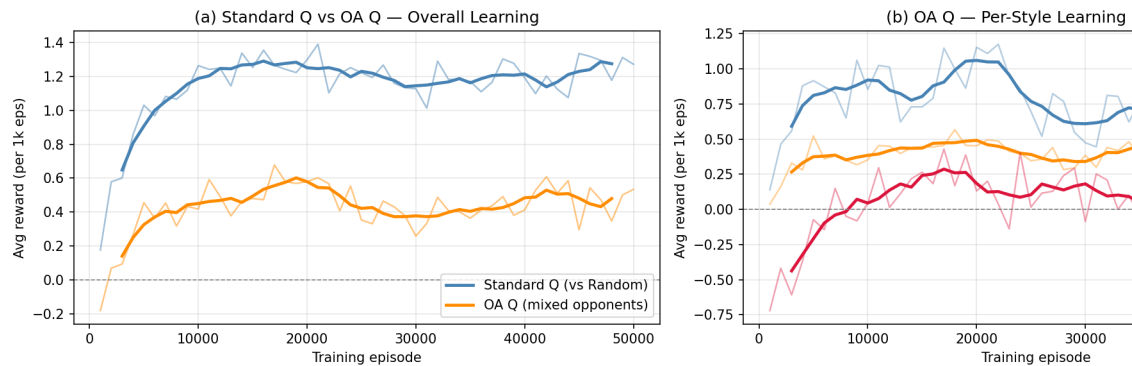


Fig. 1. Learning curves. (a) Standard Q-learning vs. Opponent-Aware Q-learning — overall training reward. (b) OA Q-learning per-style breakdown showing faster convergence against each opponent archetype.

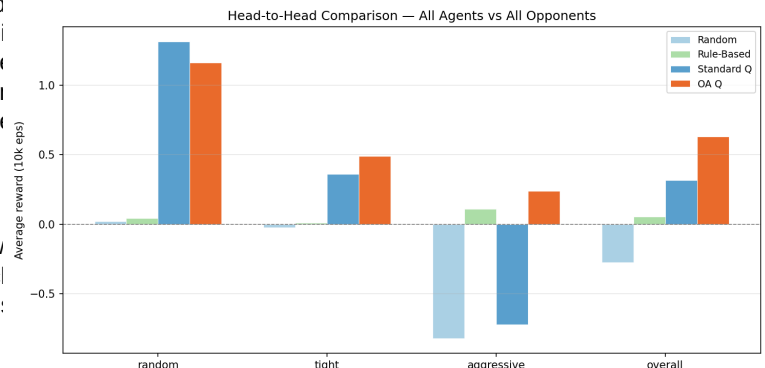


Fig. 2. Head-to-head comparison. Average reward over 10,000 evaluation episodes for all agents vs. all opponent types. OA Q-learning outperforms Standard Q-learning against both Tight and Aggressive opponents.

Fig. 1 shows that both agents achieve positive expected reward over training, but the Opponent-Aware Q-learning agent

converges more stably across all three opponent styles due to its style-conditioned Q-table partitions.

Fig. 2 confirms the central hypothesis: OA Q-learning achieves significantly higher average reward against Tight and Aggressive opponents compared to Standard Q-learning, while both agents perform similarly against the Random opponent where no exploitable pattern exists.

VII. DISCUSSION AND CONCLUSION

Our central hypothesis is that the opponent-aware agent achieves higher win rates against stylistically biased opponents compared to the style-unaware Q-learning baseline, particularly against tight and aggressive opponents where the behavioral signal is strong. Against a random opponent, both agents should perform similarly since no exploitable pattern exists.

A potential risk is that style classification is noisy at the start of an episode (before enough actions are observed), which may lead to suboptimal early decisions. Fig. 3 analyzes performance as a function of episode progress to characterize this warmup effect.

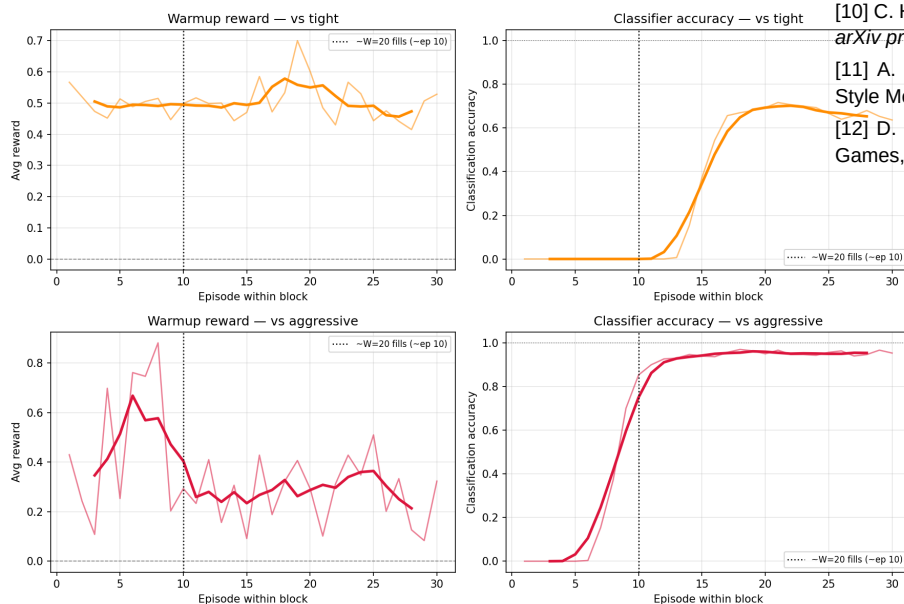


Fig. 3. Warmup analysis over 300 blocks $\times$ 30 episodes. Top: vs. Tight opponent. Bottom: vs. Aggressive opponent. Left: average reward per episode position. Right: classifier accuracy per episode position. Classification accuracy rises sharply once the $W=20$ sliding window fills (~episode 10 in block).

Fig. 3 shows that classifier accuracy rises steeply after ~10 episodes within each block (when the $W=20$ window first fills), and reward improves correspondingly. This confirms the warmup cost is bounded and short-lived.

In summary, we propose a lightweight, practical approach to opponent-aware RL in Leduc Hold'em. By using action frequency statistics rather than recurrent networks, and Q-learning rather than NFSP, our method is implementable within a course project scope while still addressing a genuine open question: does simple opponent style detection

meaningfully improve RL agent performance in a tractable poker environment? Our results confirm that it does.

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